Denoising Diffusion Models

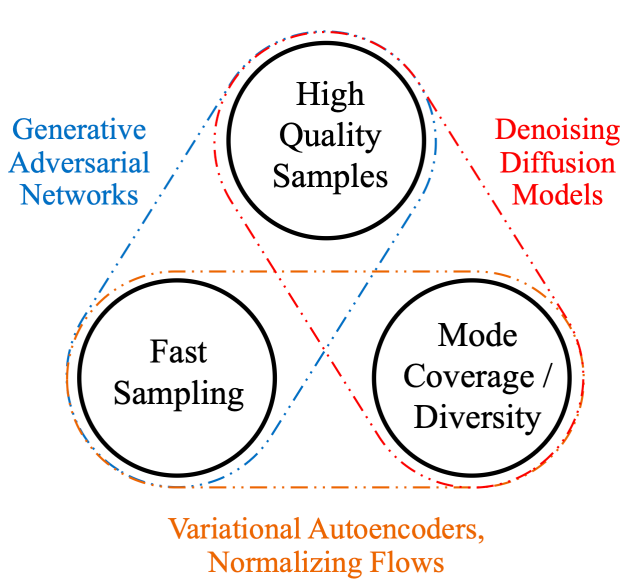
Seminar Concepts in Machine Learning | Milan Tóth | 24.09.2026

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Introduction

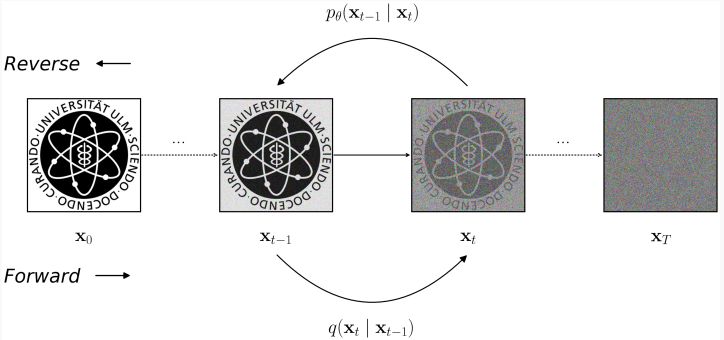
- 3 main objectives of generative image models (see Moodle)
- Trilemma: most current models achieve only 2 out of the 3
- This presentation:
    - Explain how Diffusion Models work
    - How they fit in the trilemma
    - How they try to deal with its shortcomings



Xiao, Zhisheng, Karsten Kreis, and Arash Vahdat. "Tackling the generative learning trilemma with denoising diffusion GANs." *arXiv preprint arXiv:2112.07804* (2021).

Diffusion Models

- 2 Markov Chains (Forward & Reverse)
- **Forward**: from signal to noise $q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = T_\pi(\mathbf{x}_t \mid \mathbf{x}_{t-1}; \beta_t)$
- Diffusion rates $\beta_{1 \dots T}$ can be fixed or learned
- **Reverse**: from noise to signal in small steps
- For small $\beta_{1 \dots T}$ the reverse $p(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ is of the same functional form the forward $q(\mathbf{x}_t \mid \mathbf{x}_{t-1})$
- Only Parameters of that distribution family have to be learned

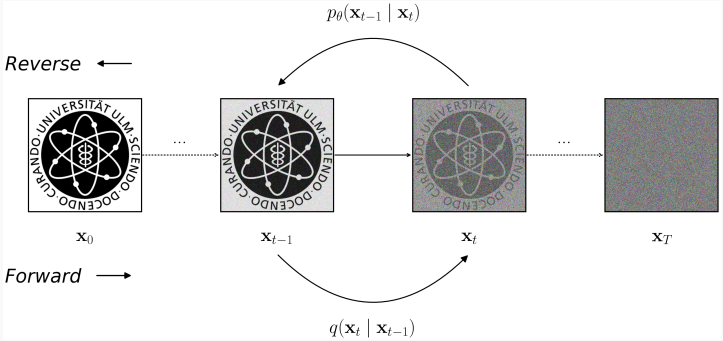


- Minimizing Lower Bound K of Log Likelihood $L = \int d\mathbf{x}_0 \, q(\mathbf{x}_0) \log p(\mathbf{x}_0)$
- $$K = -\sum_{t=2}^{T} \int d\mathbf{x}_0 \, d\mathbf{x}_t \, q(\mathbf{x}_0, \mathbf{x}_t) \, D_{\mathrm{KL}} \left(q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0) \parallel p(\mathbf{x}_{t-1} \mid \mathbf{x}_t) \right) + H_q(\mathbf{X}_T \mid \mathbf{X}_0) - H_q(\mathbf{X}_1 \mid \mathbf{X}_0) - H_p(\mathbf{X}_T)$$

Sohl-Dickstein, Jascha, et al. "Deep unsupervised learning using nonequilibrium thermodynamics." *International conference on machine learning.* pmlr, 2015.

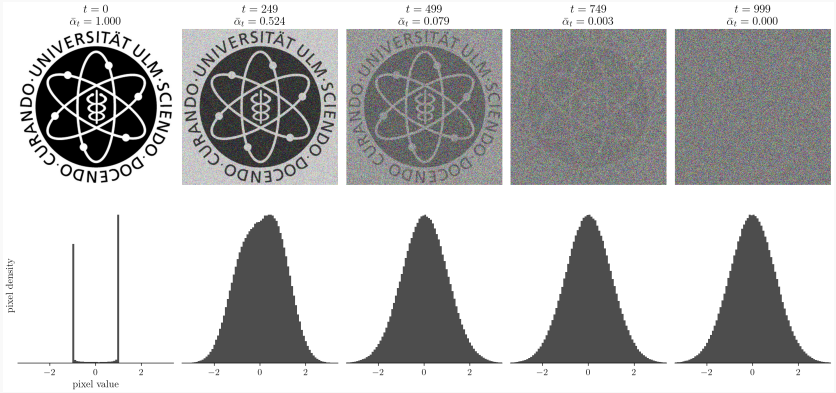
Denoising Diffusion Probabilistic Models (DDPM)

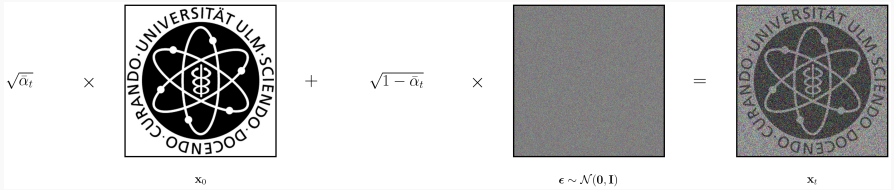
- DDPM fix the degrees of freedom of the original Framework.
- Diffusion rates $\beta_{1...T}$ are fixed and small
- $q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = T_\pi(\mathbf{x}_t \mid \mathbf{x}_{t-1}; \beta_t)$ is set to a Gaussian Kernel
- So $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ of the reverse process should be learned
- But: $\boldsymbol{\Sigma}$ is fixed, and $\boldsymbol{\mu}$ reparameterized so only the noise $\boldsymbol{\epsilon}$ added has to be learned
- K is also simplified:
- $L_{\text{simple}}(\theta) := \mathbb{E}_{t,\mathbf{x}_0,\boldsymbol{\epsilon}}\left[\left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right\|^2 \right]$



Forward Process $q(x_t \mid x_{t-1})$

- Gaussian Kernel: $q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}\left(\mathbf{x}_t;\ \mathbf{x}_{t-1}\sqrt{1-\beta_t},\ \beta_t \mathbf{I}\right),$
- With fixed β_t going linearly from $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$ in $T = 1000$ steps
- Start with clean image x_0, then noise until all that remains is Gaussian Noise x_T
- Since β_t is fixed, with $\bar{\alpha}_t := \prod_{s=1}^{t} 1 - \beta_s$, closed form: $\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

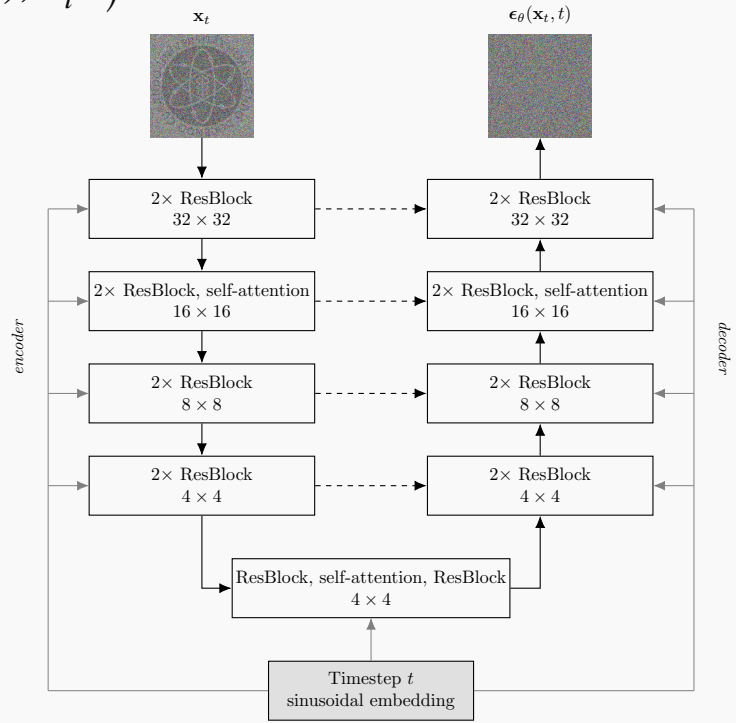




Ho, Jonathan, Ajay Jain, and Pieter Abbeel. "Denoising diffusion probabilistic models." *Advances in neural information processing systems* 33 (2020): 6840-6851.

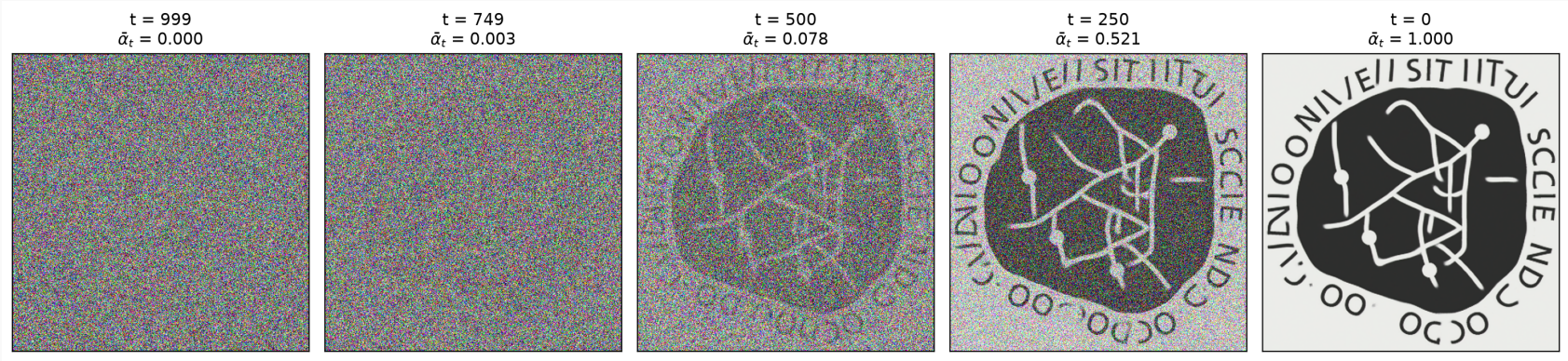
Reverse Process $p_\theta(x_{t-1} \mid x_t)$

- Gaussian reverse trajectory: $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \mathcal{N}\left(\mathbf{x}_{t-1};\, \boldsymbol{\mu}_\theta(\mathbf{x}_t, t),\, \sigma_t^2 \mathbf{I}\right)$.
- Covariance $\sigma_t^2 \mathbf{I}$ fixed to $\sigma_t^2 = \beta_t$ or $\sigma_t^2 = \tilde{\beta}_t = \dfrac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t$
- Predict added noise $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ instead of mean $\boldsymbol{\mu}_\theta(\mathbf{x}_t, t)$.
- $\boldsymbol{\mu}_\theta(\mathbf{x}_t, t) = \dfrac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \dfrac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right)$
- $\boldsymbol{\mu}_{optimal}(\mathbf{x}_t, t) = \mathbb{E}_q[\mathbf{x}_{t-1} \mid \mathbf{x}_t]$
- $\mathbf{x}_{t-1} = \boldsymbol{\mu}_\theta(\mathbf{x}_t, t) + \sigma_t \mathbf{z}, \qquad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$



Ho, Jonathan, Ajay Jain, and Pieter Abbeel. "Denoising diffusion probabilistic models." *Advances in neural information processing systems* 33 (2020): 6840-6851.

Reverse Process $p_\theta(x_{t-1} \mid x_t)$



Training and Sampling

Algorithm 1 Training

1: **repeat**
2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $t \sim \text{Uniform}(\{1, \dots, T\})$
4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: $\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\,\boldsymbol{\epsilon}$
6: Take a gradient step on $\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t,\, t) \right\|^2$
7: **until** converged

Algorithm 2 Sampling

1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \dots, 1$ **do**
3: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
4: $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$

Ho, Jonathan, Ajay Jain, and Pieter Abbeel. "Denoising diffusion probabilistic models." *Advances in neural information processing systems* 33 (2020): 6840-6851.

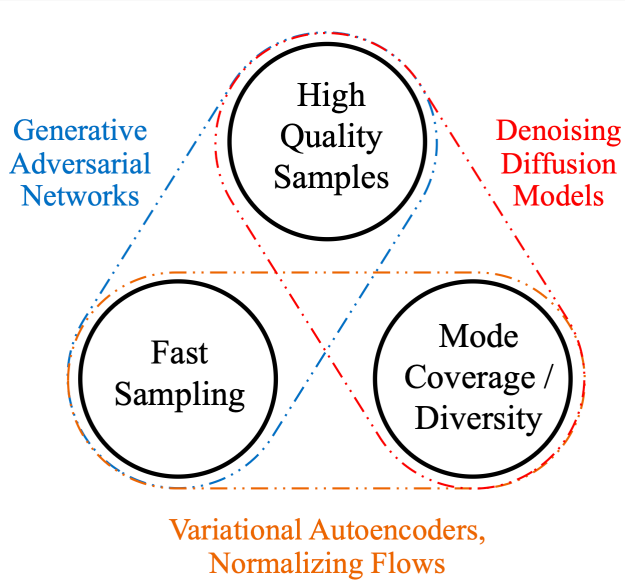
DDPM and the Trilemma

High Quality Samples:

- Empirically, yes
- A lot of small steps

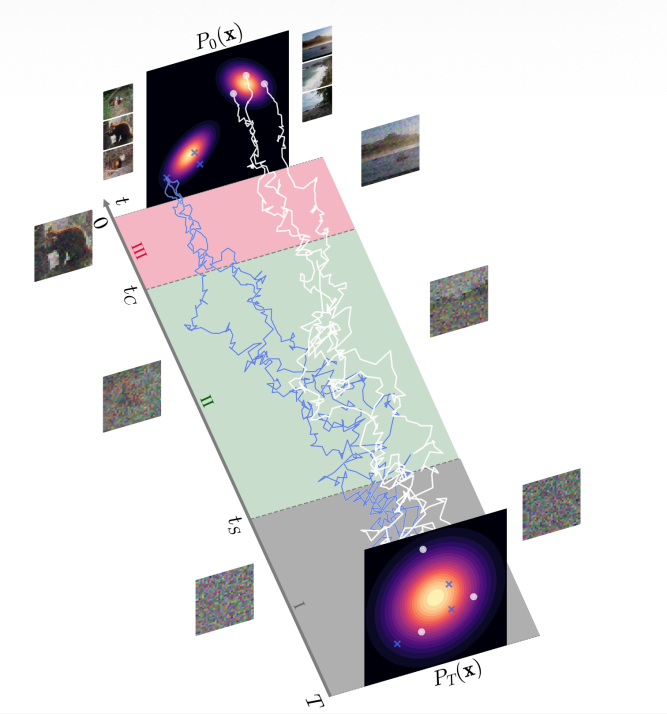
Mode Coverage:

- Empirically Good
- Objective punishes dropping modes
- But: why no blending of modes?

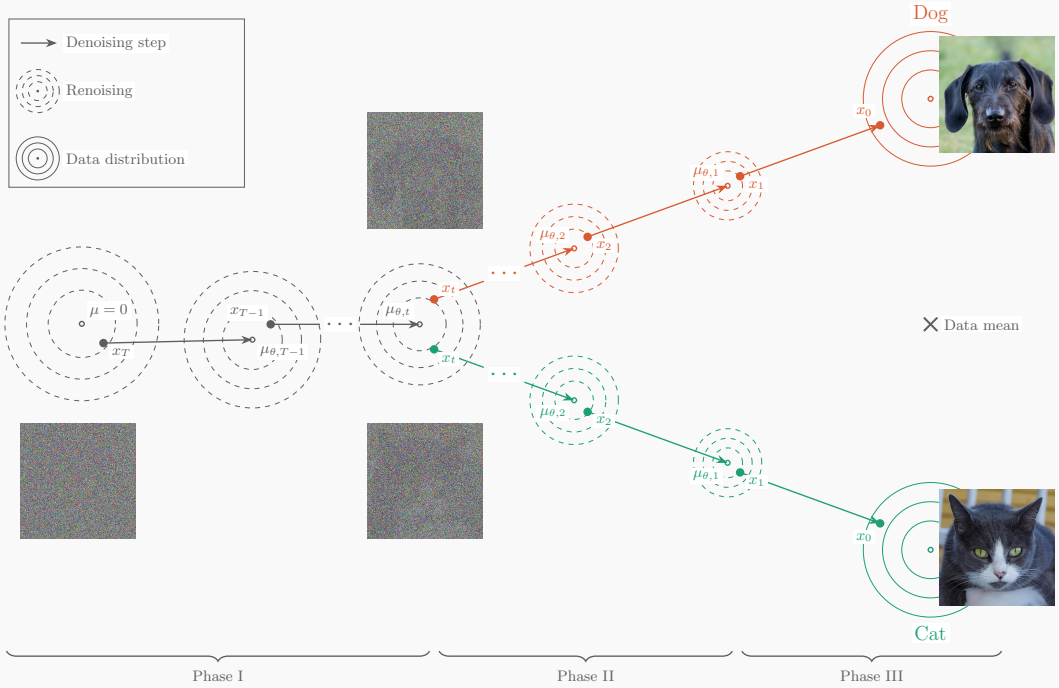


Xiao, Zhisheng, Karsten Kreis, and Arash Vahdat. "Tackling the generative learning trilemma with denoising diffusion GANs." *arXiv preprint arXiv:2112.07804* (2021).

Mode Coverage



Biroli, Giulio, et al. "Dynamical regimes of diffusion models." *Nature Communications* 15.1 (2024): 9957.



Dog and Cat images from Choi, Yunjey, et al. "Stargan v2: Diverse image synthesis for multiple domains." *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2020.

DDPM and the Trilemma

High Quality Samples:

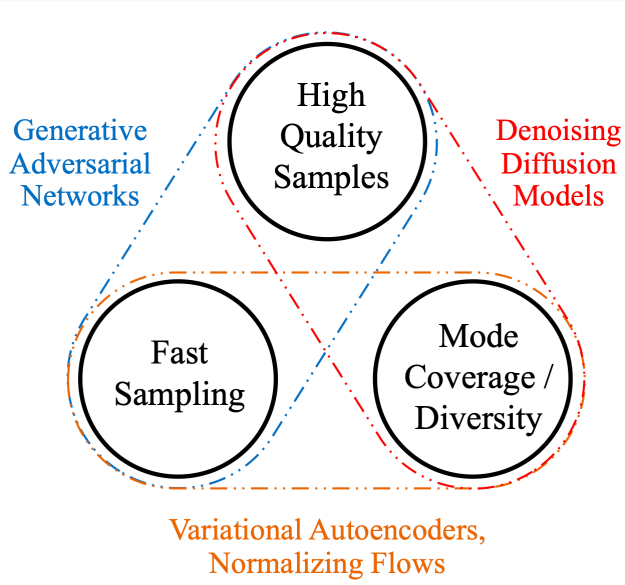
- Empirically, yes
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Mode Coverage:

- Empirically Good
- Objective punishes dropping modes
- But: why no blending of modes?

Slow Sampling:

- 50k images of size 32x32: GAN < 1 min, DDPM ~ 20h (gap widens to 1000h for 256x256)



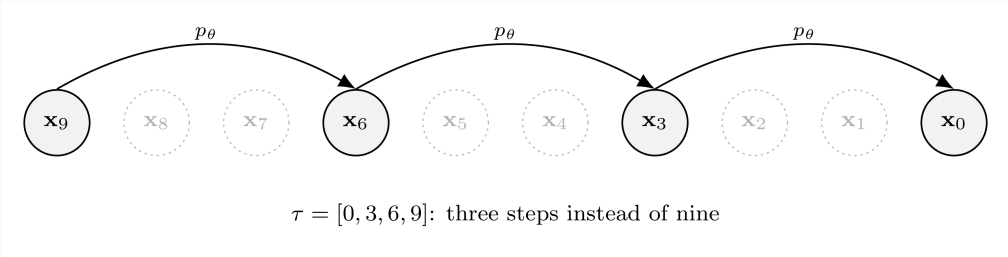
Xiao, Zhisheng, Karsten Kreis, and Arash Vahdat. "Tackling the generative learning trilemma with denoising diffusion GANs." *arXiv preprint arXiv:2112.07804* (2021).

Denoising Diffusion Implicit Models (DDIM)

- **Motivation**: Accelerating the slow sampling time of DDPMs, without changing training
- Sampling can be defined more generally with σ_t:

$$\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \underbrace{\left(\frac{\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)}{\sqrt{\bar{\alpha}_t}} \right)}_{\text{predicted } \mathbf{x}_0} + \underbrace{\sqrt{1-\bar{\alpha}_{t-1}-\sigma_t^2} \cdot \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)}_{\text{direction pointing to } \mathbf{x}_t} + \underbrace{\sigma_t \boldsymbol{\epsilon}_t}_{\text{random noise}}$$

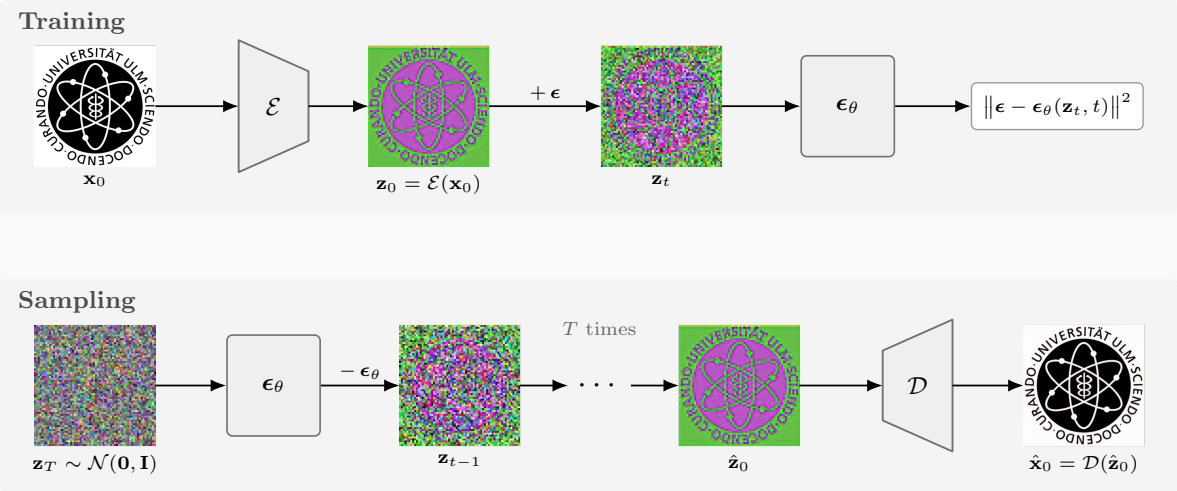
- **DDPM**: $\sigma_t^2 = \tilde{\beta}_t$; **DDIM**: $\sigma_t^2 = 0$ so deterministic
- Sample on a subsequence τ of steps (e.g. 10 of 1000); **DDIM** handles this better than **DDPM**



Song, Jiaming, Chenlin Meng, and Stefano Ermon. "Denoising diffusion implicit models." *arXiv preprint arXiv:2010.02502* (2020).

Latent Diffusion Models (LDMs)

- **Motivation:** make each reverse step cheaper, rather than reducing the number of steps
- **Key idea**: compress images.
- An Encoder maps $\mathbf{x}_0$ to a lower-dimensional latent $\mathbf{z}_0$ and a Decoder decodes once after sampling
- Cheaper per step: $\boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t)$ processes a tensor with fewer spatial positions
- Improved Quality: latent discards imperceptible detail, freeing capacity for what matters



Rombach, Robin, et al. "High-resolution image synthesis with latent diffusion models." *2022 IEEE/CVF conference on computer vision and pattern recognition (CVPR)*. ieee, 2022.

Conditioning and Guidance

Conditioning $\epsilon_\theta(\mathbf{x}_t, t, y)$:

- Network receives conditional input y

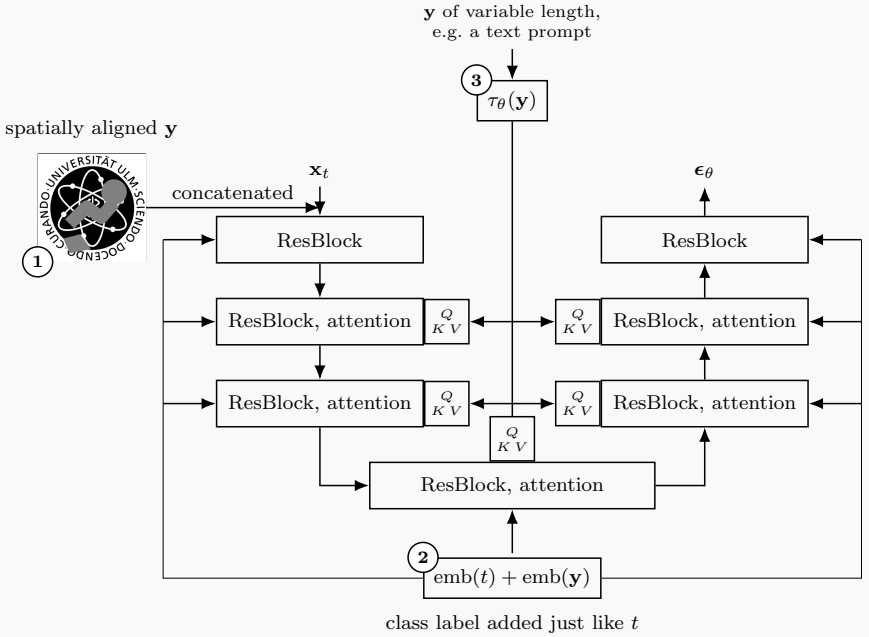
Guidance:

- Trade-off between diversity and sample quality
- Classifier Guidance:

$$\hat{\boldsymbol{\epsilon}} = \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, y) - s\sqrt{1-\bar{\alpha}_t}\,\nabla_{\mathbf{x}_t}\log p_\phi(y \mid \mathbf{x}_t),$$

- Classifier-free Guidance:

$$\tilde{\boldsymbol{\epsilon}}_\theta(\mathbf{x}_t, t, y) = (1+w)\,\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, y) - w\,\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \varnothing),$$



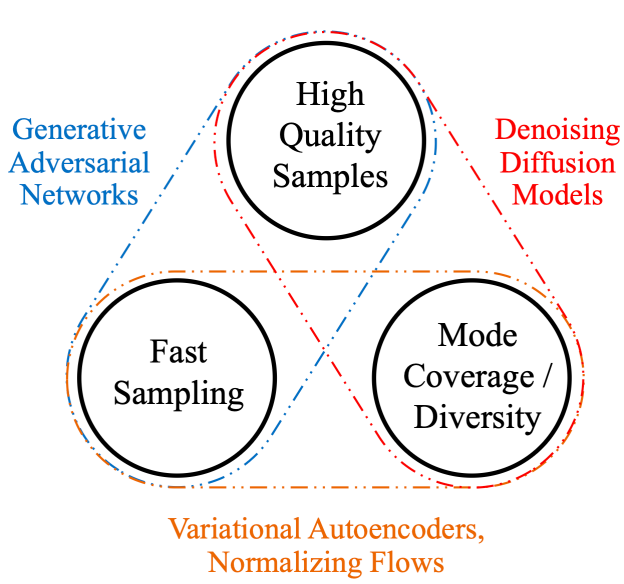
Ho, Jonathan, and Tim Salimans. "Classifier-free diffusion guidance." *arXiv preprint arXiv:2207.12598* (2022).

Dhariwal, Prafulla, and Alexander Nichol. "Diffusion models beat gans on image synthesis." *Advances in neural information processing systems* 34 (2021): 8780-8794.

Rombach, Robin, et al. "High-resolution image synthesis with latent diffusion models." *2022 IEEE/CVF conference on computer vision and pattern recognition (CVPR)*. ieee, 2022.

Conclusion

- Showed you: general Diffusion model framework, the restrictions of DDPM, Improvements of DDPM, LDM, Conditioning & Guidance
- Despite the improvements, sampling remains the bottleneck
- Trilemma remains for DDPMs



Xiao, Zhisheng, Karsten Kreis, and Arash Vahdat. "Tackling the generative learning trilemma with denoising diffusion GANs." *arXiv preprint arXiv:2112.07804* (2021).